

Computer Vision: Pedestrian Safety at SFU

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Our Problem/Users

- The problem we are working on is reducing risk for pedestrians trying to cross at marked crosswalks. Specifically we are working to improve pedestrian safety at the SFU Burnaby campus. We are doing this through detection of hazards at crosswalks via supervised learning. Hazards include “Don’t Walk” signs, vehicles, cyclists etc.
- The users/use case would be for anyone trying to navigate SFU Burnaby safely on foot. This particular model could also be very useful for the visually impaired.

Our Dataset

- We took 89 images in total around the SFU Burnaby Campus at roadways and crosswalks.
 - We got external data from the Roboflow website. This came from three separate datasets.
 - We removed the test data and created an 80/20 split for training and validation respectively.
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- Cyclist Set:
<https://universe.roboflow.com/cyclcr-vi9vn/cyclists-lt9pl>
 - Crosswalk Sign Set:
<https://universe.roboflow.com/walkassist/walk-assist/browse?queryText=&pageSize=50&startIndex=100&browseQuery=true>
 - Vehicle Set:
<https://universe.roboflow.com/faculty-of-engineering-alexandria-university-mvdnh/vehicles-sgibq>

In total we have 3910 images in our dataset.



- Crosswalks



- Cars



- Don't Walk
- Traffic Signal



- Walk
- Traffic Signal



- Bicycle



- Bus

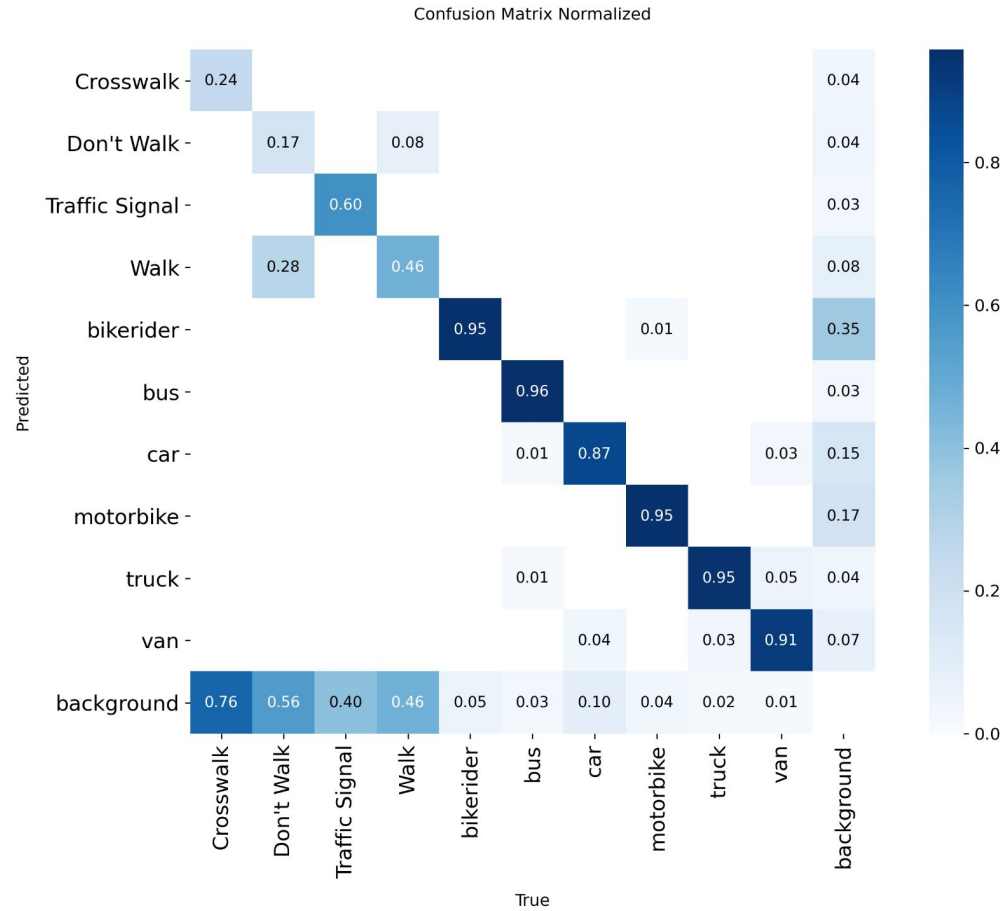
Challenges When Collecting Data

- We had to wait around for cars to drive by during data collection.
- It was hard to find cyclists to take pictures of.
- While looking for external data it was hard to find images that matched how crosswalks on the Burnaby campus looked. (ex. Instead of a hand signal for “Do not cross” we were finding signs that had pedestrian symbols or the words “Do Not Cross”)

Biases/Issues with Dataset

- One of the external datasets we used contained vehicles that did not exist on Burnaby campus (ex. threewheel). We resolved this issue by removing the category all together, including the images that were in the category.
- Our data was mainly collected on a sunny day, meaning our data may be biased towards nicer weather. We tried to fix this by collecting data on two different days.

Initial Results of Training



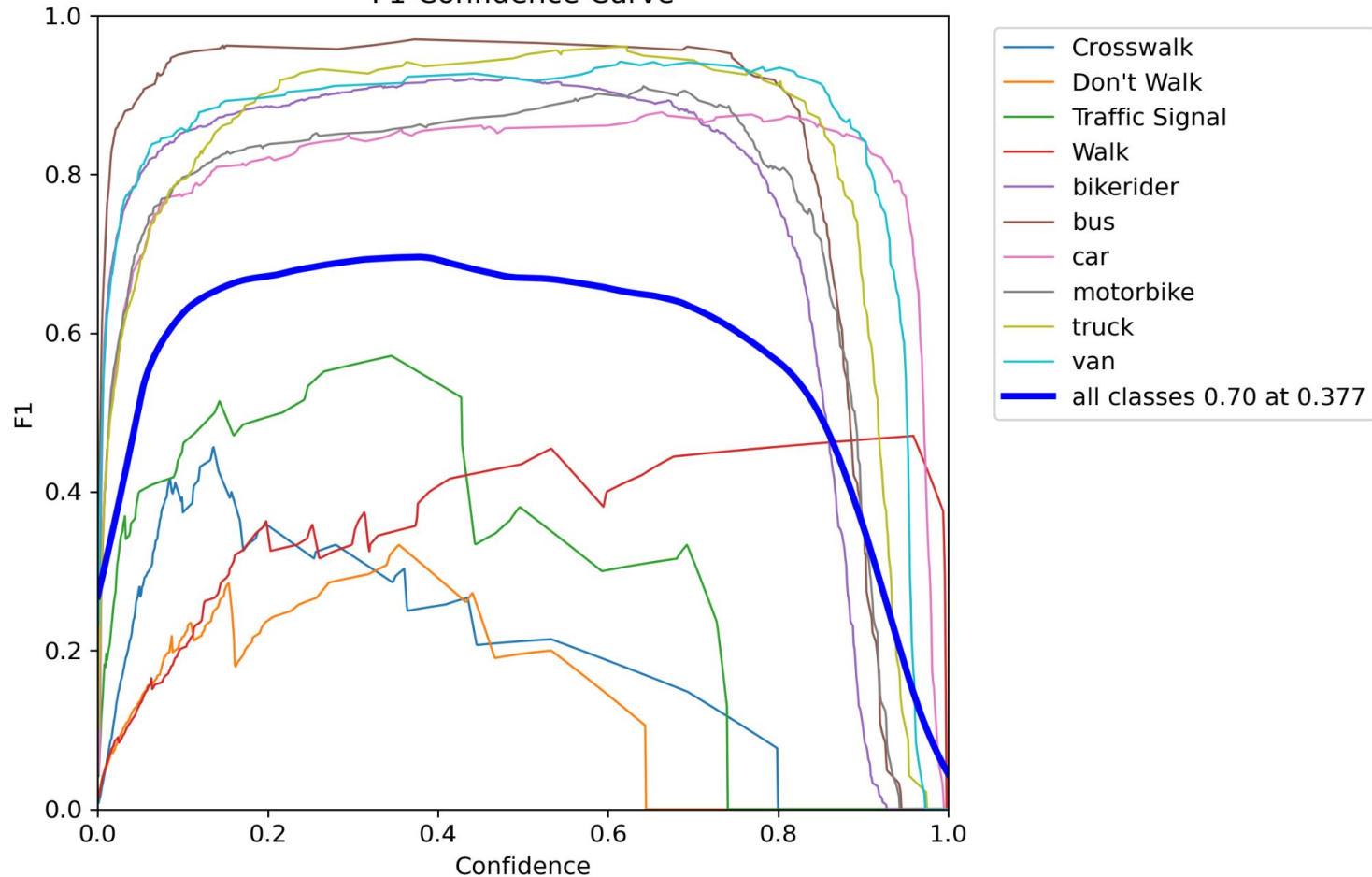
```

Ultralytics 8.3.221 🚀 Python-3.12.12 torch-2.8.0+cu126 CUDA:0 (Tesla T4, 15095MiB)
Model summary (fused): 72 layers, 11,129,454 parameters, 0 gradients, 28.5 GFLOPs
val: Fast image access ✅ (ping: 0.6±0.2 ms, read: 22.1±14.6 MB/s, size: 38.7 KB)
val: Scanning /content/drive/MyDrive/CV Project Oct 24/labels/val.cache... 788 images, 0 backgrounds, 0 corrupt: 100%
      Class      Images  Instances   Box(P          R      mAP50  mAP50-95): 100% _____ 50/50 3.1:
        all         788       1080      0.76      0.684      0.721      0.584
      Crosswalk         22         25      0.602      0.16      0.341      0.168
      Don't Walk         16         18      0.645      0.204      0.251      0.08
    Traffic Signal         11         15      0.603      0.507      0.47      0.291
          Walk         13         13      0.39      0.385      0.425      0.293
      bikerider        251        414      0.897      0.935      0.962      0.709
          bus         103        119      0.983      0.958      0.977      0.918
          car         125        134      0.849      0.858      0.903      0.871
      motorbike         104        137      0.824      0.92      0.932      0.712
          truck          75         92      0.917      0.96      0.978      0.867
          van         103        113      0.894      0.956      0.974      0.935

Speed: 1.1ms preprocess, 7.7ms inference, 0.0ms loss, 1.6ms postprocess per image
Results saved to /content/runs/detect/val
np.float64(0.6841793578489189)

```

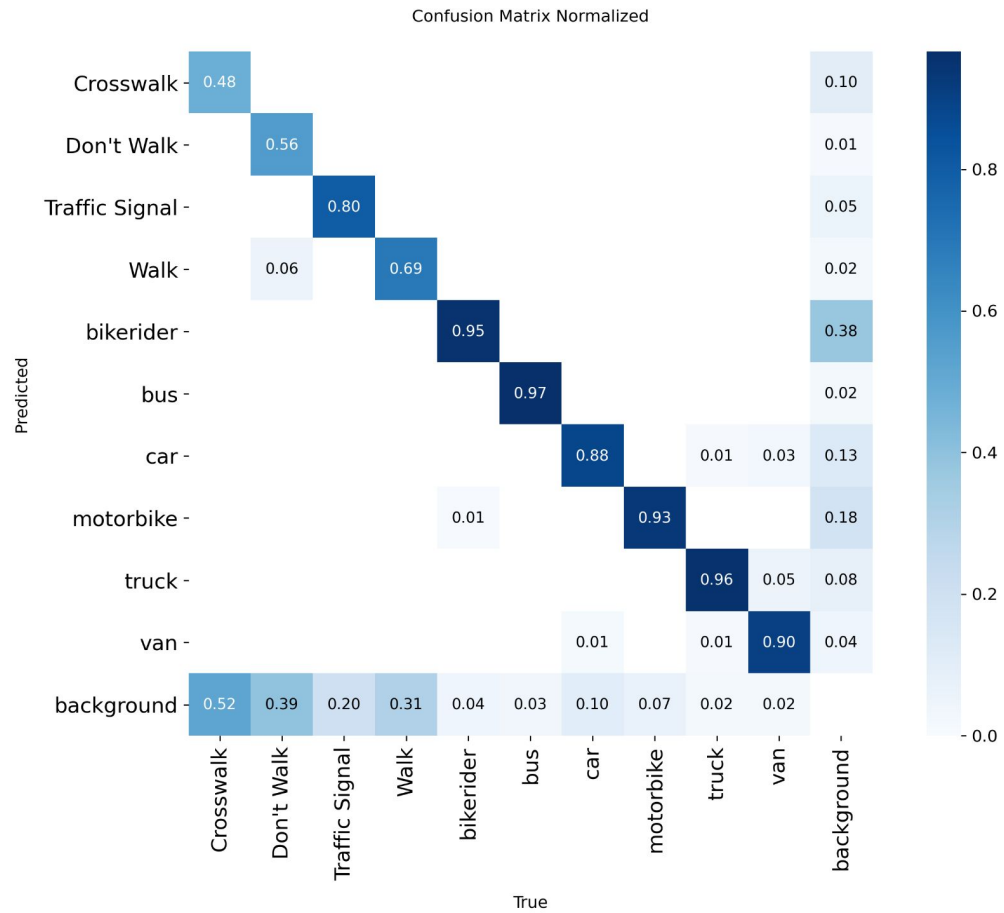
F1-Confidence Curve



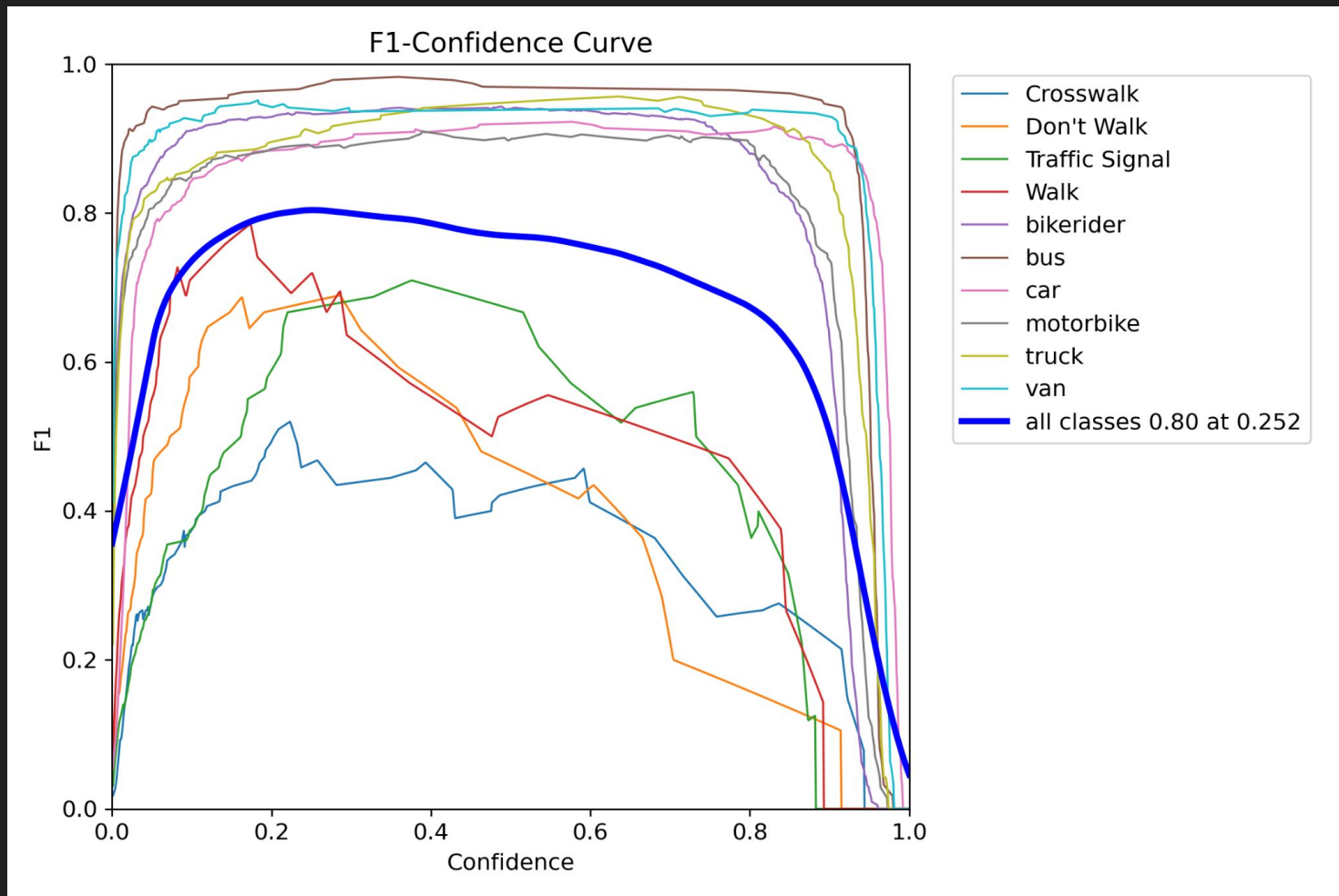
Results After Hyperparameter Change

Epoch 5 -> Epoch 15

Patience 5 -> Patience 15



Class	Images	Instances	Box(P	R	mAP50	mAP50-95):
all	788	1080	0.82	0.803	0.812	0.658
Crosswalk	22	25	0.499	0.44	0.435	0.181
Don't Walk	16	18	0.884	0.556	0.584	0.337
Traffic Signal	11	15	0.622	0.733	0.544	0.33
Walk	13	13	0.746	0.68	0.788	0.515
bikerider	251	414	0.912	0.954	0.968	0.767
bus	103	119	0.978	0.966	0.988	0.94
car	125	134	0.888	0.896	0.926	0.895
motorbike	104	137	0.852	0.927	0.951	0.776
truck	75	92	0.863	0.957	0.949	0.875
van	103	113	0.954	0.924	0.987	0.96



Images that Worked



Images That Did Not Work



Biases and Possible Ethical Issues

- Our biggest potential ethical issue is our model classifying instances of “Walk” and “Don’t Walk” in the same image. We did not anticipate this being a problem but it can put pedestrians in danger.
- The datasets we used are public and open source so there should be no ethical issues regarding ownership.

Circumstances of Success and Failure

- Our model may excel at detecting certain hazards for pedestrians, as it is accurate in predicting vehicles due to the extensive and diverse vehicle dataset we used. This is important because vehicles pose the most imminent threat to walking pedestrians.
- The model seems to fail most when detecting “Walk” and “Don’t Walk” signs. We think this could be fixed with a more diverse dataset of these signs.